

Do Latent World Model Representations Exhibit Physical Expectation?

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Abstract

A central goal of cognitive science and machine learning is understanding whether learned representations support predictive internal models: compact structures that anticipate how the world evolves under action. Violation-of-expectation (VoE) paradigms, borrowed from developmental psychology, interpret increased prediction error under physically implausible outcomes as evidence of expectation-like behavior. We critically examine this assumption by evaluating predictive world-model representations, pretrained self-supervised visual representations, and reconstruction-based representations across three environments. We find that naive VoE signals are largely explained by latent geometry: prediction error increases with distance in representation space, not necessarily with structured violations of physical plausibility. When latent distance is controlled, the violation signal is reduced to near baseline. Prediction error is further only weakly sensitive to action perturbations under our latent-predictor probe, indicating limited evidence for action-conditioned dynamics in this evaluation protocol. These results suggest that VoE evaluations can conflate smooth representational structure with predictive expectation, motivating latent-distance and action-dependence controls for rigorous world-model evaluation.

Introduction

A long-standing question in cognitive science and machine learning is whether learned representations support predictive internal models of the world. In Craik’s formulation, thought consists of constructing internal representations that anticipate the consequences of external processes [2]. These representations are not merely descriptive: they must support transformations that mirror the causal structure of the environment. Tolman similarly argued that behavior is guided by structured internal representations rather than simple stimulus-response associations [15]. In both accounts, intelligence depends not only on encoding the world, but on representing how it evolves under action [9].

Recent work on latent world models, particularly action-conditioned Joint Embedding Predictive Architectures (JEPAs), pursues this idea computationally by learning abstract representations in which future states can be predicted under candidate actions. In the JEPA view, the goal is not to reproduce every sensory detail, but to preserve the aspects of state that are predictable and useful for action. This contrasts with reconstruction-based methods, which are trained to preserve visual information but are not directly optimized to learn action-conditioned transition structure.

Despite this shift, evaluating whether learned representa-

tions actually support predictive structure remains difficult. A common approach, borrowed from developmental psychology, uses violation-of-expectation (VoE) paradigms: increased prediction error under physically implausible outcomes is interpreted as evidence of learned expectations. However, such signals may also arise from smooth latent geometry, where physically distant states are farther apart in representation space, rather than from an action-conditioned model of physically plausible transitions.

We investigate this possibility directly. We ask: once a representation has been learned, does it support structured, action-conditioned prediction, or does a VoE-style score primarily reflect geometric similarity? In our experiments, naive VoE effects are largely explained by latent geometry, are reduced to near baseline under distance-matching controls, and are only weakly sensitive to action perturbations under a Ridge latent-predictor probe. This suggests that VoE-style evaluations of learned world-model representations require controls that separate predictive expectation from representational smoothness.

Contributions

- We introduce a controlled evaluation framework for testing whether learned representations provide evidence of predictive, action-conditioned structure across multiple environments.
- We show that naive violation-of-expectation (VoE) signals can be largely explained by latent geometry rather than by evidence of structured physical expectation alone.
- We design controls that separately test the roles of latent distance and action dependence in prediction error.
- We show that reconstruction-based and pretrained self-supervised visual representations can exhibit apparent VoE effects, suggesting that VoE signals do not by themselves establish action-conditioned world modeling.

Background

Latent World Models and Representation Learning

Recent approaches to latent world modeling train models to predict future representations conditioned on context and action, without requiring reconstruction of raw sensory input [10]. This framework, exemplified by action-conditioned Joint Embedding Predictive Architectures (JEPAs), aims to learn abstract state representations that preserve predictable,

action-relevant structure while discarding details that are irrelevant or inherently unpredictable (e.g., pixel value of a leaf in the wind). It contrasts with reconstruction-based methods such as masked autoencoders, which are trained to reconstruct masked image content but do not explicitly encourage action-conditioned dynamics, and with pretrained self-supervised visual models, which often organize images by semantic or perceptual similarity but are not necessarily trained to model temporal evolution.

These differences correspond to distinct *inductive biases*: objective-level and architectural pressures that shape what information a representation tends to preserve. We identify three relevant dimensions. *Rendering bias* concerns whether the model is trained to reconstruct raw observations. Pixel reconstruction encourages the latent state to preserve appearance-level information such as texture, lighting, and background, even when these factors are not central to the dynamics. Non-reconstructive predictive objectives face less direct pressure to preserve such detail; their representations are instead shaped by what is useful for prediction.

Representational prior concerns what structure the encoder brings to the problem: DINOv2 inherits semantic and perceptual features from large-scale self-supervised visual pre-training, carrying strong biases toward object identity, texture, and spatial layout. By contrast, LeWM’s encoder is trained in-domain from scratch under a predictive objective and anti-collapse regularization, importing far less external visual prior.

Temporal scope concerns whether the model is pressured to represent information useful across multiple time horizons. In the world-model view, the relevant state is not an objective copy of the environment, but an abstraction suited to the prediction horizon and control problem. Thus, different training objectives embody different implicit theories of what a useful state representation is for.

Our goal is not to reject the world-model framing, but to test whether a common representation-level VoE protocol actually detects the action-conditioned predictive structure that world models are meant to provide.

Prediction, Causality, and Internal Models

Craik’s theory of thought frames prediction as the manipulation of internal structures corresponding to external processes [2]. For a representation to support prediction in this sense, it must encode not only the current state of the world but also the transformations induced by actions. This aligns with causal theories of explanation, in which understanding involves modeling how interventions produce changes in outcomes [8].

Purely relational representations, by contrast, may capture statistical associations between observations without modeling the processes that generate transitions between them. Such representations may support similarity judgments or interpolation but do not necessarily enable structured, action-conditioned prediction. The distinction matters for evaluation: a system can exhibit apparently structured behavior, such as

increased prediction error for physically distant states, without modeling causal processes at all.

Cognitive Maps and Structured Representation

Tolman’s cognitive maps offer a complementary framing: structured internal representations that support flexible navigation and goal-directed behavior by encoding relationships between states rather than merely embedding observations in a smooth space [15]. The idea that cognition depends on structured, learnable concepts rather than raw perceptual associations runs through much of cognitive developmental research [1]. In computational terms, the closest formal analog is the bisimulation metric [4, 3], originally developed to formalize state abstraction in Markov Decision Processes. Two states are bisimilar when they yield identical expected reward distributions under all action sequences. A bisimulation-based representation clusters perceptually distinct states together when they are behaviorally indistinguishable, and separates perceptually similar states when they have different causal consequences.

Cognitive theories predict this selectivity: abstract over perceptual noise, preserve causal structure. Critically, a bisimulation-style representation is dynamics-aware: it preserves distinctions between states when those distinctions change future transition behavior under actions, while abstracting over differences that do not affect future outcomes. This makes it a useful formal analogy for asking whether a representation supports flexible planning rather than merely perceptual similarity.

The Slow-Features Problem as a Cognitive Analog

A documented failure mode of pretrained visual encoders in world modeling illuminates the cognitive relevance of inductive bias. Because DINOv2 was trained to distinguish visual scenes, its features are sensitive to slowly-changing visual properties such as background texture. In a planning environment, the background does not change during a trajectory, so it is correlated with every state; the predictor learns to exploit it. When the background changes at test time, the latent space is no longer meaningful, and planning performance degrades substantially [11].

This is exactly what a bisimulation-based cognitive theory predicts should not happen: background changes are task-irrelevant, so a good planning representation should be invariant to them. The failure is a direct consequence of the representational prior inherited from pretraining on visual scene discrimination rather than causal state modeling. Human perception, by contrast, is tuned to action-relevant features rather than maximally visually distinctive ones [15]. LeWM, which has no pretrained encoder and no reconstruction pressure, may avoid this failure by organizing its latent space around whatever is predictively useful. This should, in a controlled environment, be causal state. Whether this is empirically the case is one of the questions our geometry experiments attempt to address.

Evaluating Representations via Alignment and Expectation

Representational alignment methods measure correspondence between learned latent distances and human similarity judgments or neural data ([13]; [14]). While informative, alignment alone does not establish that a system supports the same computations as humans. A model can encode relational similarity without encoding the causal processes that generate transitions. Violation-of-expectation paradigms extend this idea by probing predictive structure. In developmental psychology, VoE experiments infer the presence of internal expectations by measuring increased surprise when observed outcomes violate physical constraints [5, 6, 16]. Computational accounts of curiosity and information-seeking similarly treat surprise as a signal that drives internal model updating [7]. In machine learning, analogous evaluations measure prediction error under mismatched or counterfactual futures, interpreting higher error as evidence that the model has learned to expect physically plausible outcomes. This interpretation assumes, however, that prediction error reflects action-conditioned models of the world rather than generic properties of representational geometry. This is an assumption our experiments directly test.

Methods

Environments

We evaluate representations across three environments that vary in their dynamical structure, observability, and task complexity:

- **PushT** is a planar manipulation task in which an agent must push a block to a target configuration. The ground-truth variables include *agent and block position, orientation, and velocity, introducing non-trivial contact dynamics* in which useful representations may need to capture relationships between multiple entities.
- **TwoRoom** is a 2D navigation task in which an agent moves between two spatial regions separated by a barrier. The state is the agent’s 2D position. Because the underlying state is low-dimensional and fully observable, TwoRoom provides a setting in which representation geometry can be cleanly interpreted, aligning closely with classical cognitive map paradigms [15].
- **Reacher** is a continuous control task in which a two-link robotic arm must reach a target position. Observations include joint configurations and task-relevant features such as fingertip and target positions. Unlike PushT, the dynamics are governed by smooth continuous control rather than contact interactions.

Together, these environments span a range of representational demands: structured object interaction (PushT), spatial abstraction (TwoRoom), and continuous kinematics (Reacher). This diversity allows us to evaluate whether observed effects generalize across settings rather than arising from a specific environment.

Representations

We evaluate four representation classes corresponding to distinct training paradigms:

- **LeWM** is a predictive latent world model trained with a joint-embedding predictive objective [12]. It learns an encoder and an action-conditioned predictor so that future latent states can be predicted in representation space without requiring pixel reconstruction.
- **DINOv2** is a pretrained self-supervised vision model trained by self-distillation. In our experiments, it provides visual embeddings that capture semantic and perceptual structure, but it is not itself trained to model action-conditioned temporal dynamics.
- **MAE** is a masked autoencoder trained to reconstruct masked image content; its objective preserves visual information but does not explicitly require modeling how states evolve under actions.
- **Random** features, matched in dimensionality, provide a baseline for effects arising from dimensionality and generic distance structure rather than learned visual or physical structure.

Experimental Framework

We evaluate representations using a sequence of experiments that isolate what is encoded, how it is organized, and whether it provides evidence of predictive expectation. Let x_t denote observations, a_t actions, and $z_t = \phi(x_t)$ the latent representation from encoder ϕ .

Experiment 1: Representation Probing

We train linear probes g (RidgeCV, $\alpha \in [10^{-4}, 10^4]$) to predict ground-truth state variables from latent embeddings: $\hat{s}_t = g(z_t)$, using an 80/20 train/test split. We report R^2 per state dimension and averaged across all dimensions. This establishes what physical variables are encoded, though high decodability does not imply structured prediction.

Experiment 2: Latent Geometry

We measure Spearman rank correlation between pairwise latent distances $d_{\text{lat}}(z_i, z_j)$ and pairwise physical distances $d_{\text{phys}}(s_i, s_j)$, testing whether latent space is organized like a cognitive map of the environment. We also report a causal-to-perceptual alignment ratio,

$$\text{CPAR} = \frac{\rho_{\text{causal}}}{\rho_{\text{pixel}}},$$

where ρ_{causal} measures alignment between latent distances and task-relevant physical distances, while ρ_{pixel} measures alignment between latent distances and pixel-space distances. Values above 1 indicate that latent geometry is more aligned with task-relevant state variables than with raw visual similarity.

We report correlations against multiple physical targets per environment, such as block position, agent position, and visible configuration in PushT.

Experiment 3: Violation-of-Expectation (VoE)

A Ridge regression predictor f is trained on windows of latent history ($k = 3$ frames) concatenated with the corresponding action block ($h = 5$ steps) to predict the next latent state: $\hat{z}_{t+h} = f(z_{t-k:t}, a_{t:t+h})$. Surprise is measured as $S = \|\hat{z}_{t+h} - z_{\text{target}}\|_2$ under four target conditions: the true next state (normal), a physically nearby state (near, 45th–55th percentile of physical distance), a physically distant state (far, above 90th percentile), and a random state drawn uniformly from the dataset. The pattern $S_{\text{normal}} < S_{\text{near}} < S_{\text{far}}$ is often interpreted as a signature of expectation-like behavior; we also report the ratio `far_over_normal` as our primary metric.

Control 1: Latent-Distance Matching

For each target pair $(z_{t+h}, z_{\text{target}})$, we select a comparison target $\tilde{z}$ satisfying

$$\|z_{t+h} - \tilde{z}\|_2 \approx \|z_{t+h} - z_{\text{target}}\|_2$$

while varying physical distance. If the VoE effect is driven by physical structure beyond latent distance, it should persist; if it is driven primarily by geometry, it should move toward 1.0.

Control 2: Action Perturbation

We replace the true action sequence with a shuffled sequence $\tilde{a}_{t:t+h}$ and compute the sensitivity ratio

$$R_{\text{action}} = \frac{\mathbb{E}[\|\hat{z}_{t+h}^{\text{shuffle}} - z_{t+h}\|_2]}{\mathbb{E}[\|\hat{z}_{t+h} - z_{t+h}\|_2]}.$$

Values near 1 indicate weak sensitivity to action perturbations under this probe, suggesting that prediction may be driven more by latent history than by the supplied action sequence.

Results

Representation Geometry

Linear probes (RidgeCV) reveal a consistent pattern across all three environments: spatial state variables are well-encoded by every learned representation, while velocity is not recoverable by any model. In PushT, block position and angle are decoded with $R^2 \geq 0.97$ by LeWM, DINOv2, and MAE alike (e.g., `block_x` $R^2 = 0.998, 0.988, 0.998$ respectively under that random policy). Agent velocity yields $R^2 < 0$ for all models in all environments, a results that holds in PushT under a nonlinear MLP probe, ruling out a linearity artifact. The three learned architectures are indistinguishable on probing accuracy, indicating that spatial content reflects the environment’s static frames rather than architectural differences. Latent geometry results are more discriminating. In PushT, LeWM’s latent space is most tightly organized around the task-relevant causal object: Spearman ρ between latent distance and block-position distance reaches 0.649 (`pusht_weak`) and 0.672 (`pusht_random`), compared to 0.424 and 0.321 for DINOv2, and 0.489 and 0.488 for MAE. Notably, DINOv2’s latent geometry tracks visual configuration ($\rho = 0.463$) more than block position ($\rho = 0.321$) under the random policy – the only model for which the causal-to-perceptual alignment

ratio ($\text{CPAR} = \rho_{\text{causal}} / \rho_{\text{pixel}}$) falls below 1.0 ($\text{CPAR} = 0.694$). LeWM achieves $\text{CPAR} = 1.394$ in the same setting; MAE achieves $\text{CPAR} = 1.156$.

Naive Violation-of-Expectation

Using a Ridge predictor trained on latent history and action sequences, naive `far_over_normal` ratios are large across all learned models and environments: LeWM reached $5.42\times$ in `pusht_random`, MAE $5.08\times$, DINOv2 $4.22\times$. Under the weak policy the pattern is similar (LeWM $2.68\times$, MAE $1.82\times$, DINOv2 $1.64\times$). The random baseline yields ratios ≤ 1.00 throughout. Taken at face value, these results suggest all learned representations exhibit graded sensitivity to physical plausibility.

Latent-Distance Control

The naive VoE interpretation is strongly weakened by the latent-distance matching control. When violation targets are selected to match the latent distance from the true next state, holding representational distance approximately constant while varying physical distance, the `far_over_normal` ratio is reduced to near baseline. Across all models and environments, the controlled ratio falls between 0.986 and 1.065 (Table 1), compared to naive effects reaching up to $5.4\times$ in `pusht_random`. This suggests that the dominant driver of the measured VoE signal is latent geometry: physically distant states tend to be farther in latent space, so prediction error can appear to track physical plausibility even without an independent action-conditioned expectation signal.

Action Dependence

Replacing the true action sequence with a shuffled sequence changes prediction error by at most 12% (LeWM, `tworoom_expert`) and by less than 3% in the primary PushT environments. For DINOv2 and MAE, the penalty is consistently below 1% across all runs. A complementary no-action control run on Reacher (the only environment where this condition was evaluated) shows that removing actions entirely *increases* the naive VoE effect (LeWM: $1.59\times \rightarrow 4.22\times$; DINOv2: $1.37\times \rightarrow 2.08\times$; MAE: $1.39\times \rightarrow 2.29\times$). Under this Ridge-predictor probe, these results provide limited evidence that the measured VoE signal depends on action-conditioned dynamics. Instead, actions appear to contribute little beyond latent history in this evaluation, and in some cases may add noise to a predictor that is primarily exploiting latent smoothness.

Summary of Controls

Together, the controls show that the apparent VoE signal has two properties that weaken the predictive-expectation interpretation: it is largely explained by latent geometry in Control 1, and it is only weakly affected by action information under our probe from Control 2. The reduction toward baseline is consistent across environments and representation families.

Discussion:

Apparent Violation-of-Expectation as a Geometric Artifact

Our central result is that naive VoE evaluations, when applied to learned latent representations through our Ridge-predictor protocol, can overstate evidence for physical expectation. The issue is not the world-model objective itself: in the LeCun/JEPA framing, world models are meant to be action-conditioned predictors over abstract state representations. Rather, the issue is that a VoE score can reflect representational geometry unless the evaluation controls for latent distance and action dependence. This conclusion is supported by the following empirical results:

- From the results of [Control 1](#), prediction error increases for physically distant targets. This is consistent with a property of the latent space, such that, smooth geometric structure where distance correlates with physical dissimilarity, rather than a learned model of what futures are actually physically plausible. We can recognize this as a well-known confound in cognitive neuroscience: representational similarity analyses can reflect low-level feature overlap rather than the structured representations they are used to infer.
- The effect is reduced to near baseline under latent-distance matching. This does not rule out expectation-like structure in all possible probes or models, but it does show that our measured VoE signal is largely accounted for by latent geometry.
- A system implementing expectation in the sense of [2] or [15] should encode not just what states look like, but how they are reached. In [Experiment 3](#), the Ridge predictor can produce a strong naive VoE signal even when actions are shuffled or removed, suggesting that it is often acting more like a smooth latent interpolator than a genuinely action-conditioned dynamics model. This does not show that action-conditioned world models cannot learn such structure; rather, it shows that observation-level similarity alone is insufficient evidence for the kind of predictive internal model invoked by cognitive theories of expectation.

Limits of Action-Conditioned Prediction

The weak action dependence observed under our Ridge-predictor probe is not specific to one representation family. Specifically, for LeWM representations, a Ridge predictor trained on latent histories and actions demonstrates only a 2.4% action-shuffle penalty in **PushT**, despite LeWM itself being trained with an action-conditioned latent prediction objective. This may reflect a limitation of the linear Ridge predictor, rather than the representation itself, as the probe may be insufficiently expressive to extract action-conditioned structure that exists in the representation. A nonlinear predictor, or one trained specifically on the downstream planning task, may reveal stronger action-dependence. Thus, we consider this as an open question rather than a definitive negative result about any model in particular.

More broadly, the failure suggests that the representational properties measured by probing and geometry in [Experiment 1](#) and [Experiment 2](#), which are genuinely discriminating between models, do not straightforwardly translate into action-conditioned dynamics. Analysis of the latent-space geometry is a necessary, but not sufficient, condition for the kind of internal models that cognitive theory describes.

Implications for Evaluating World Models

The results and observations motivate two methodological conclusions:

- VoE-style evaluations should not be reported without latent-distance and action-shuffle controls. Without these, the effect size reflects the tightness of the model’s latent geometry rather than the sophistication of its predictions.
- The latent geometry measures from [Experiment 2](#), particularly the CPAR metric, appear more diagnostically useful than naive VoE for distinguishing architectures. DINOv2’s sub-unity CPAR in **PushT** is consistent with the slow-features failure mode described in recent work on planning with pretrained visual encoders [11, 17], while LeWM’s higher CPAR aligns with its training objective of next-state prediction without reconstruction pressure.

From a cognitive science perspective, our results reinforce a distinction that Craik’s original account already drew: a system can encode relational similarity between states (which all our learned models do, to varying degrees) without encoding the causal processes that generate transitions between states. The former is what our probing and geometry experiments measure. The latter is what VoE evaluations are meant to measure, but in practice do not, given current evaluation designs. [2]

Conclusion

We have shown that, in our representation-level VoE protocol, naive violation-of-expectation effects are largely explained by latent geometry rather than by predictive expectation alone. Prediction error increases for physically distant targets, is reduced to near baseline when latent distance is controlled, and is only weakly affected by whether action sequences are correct, shuffled, or absent under our probe. This pattern holds across the three learned representation families we evaluate: LeWM, DINOv2, and MAE.

These results do not reject the world-model framing. Rather, they clarify what a rigorous evaluation of world-model representations must control for. In the JEPA view, a world model should be an action-conditioned predictor over abstract state representations, not a pixel-level simulator. A VoE score should therefore provide evidence of action-conditioned predictive structure, not merely smooth latent geometry. Latent geometry, measured by the causal-to-perceptual alignment ratio, provides a more discriminating and interpretable complement to VoE because it distinguishes LeWM from DINOv2 in the direction suggested by their training objectives without

conflating smoothness with expectation. We therefore recommend latent-distance and action-shuffle controls as a minimal standard for VoE-style evaluations of learned world-model representations.

Table

Model	Naive	Controlled	R_{action}
LeWM	2.63	1.017	1.043
DINOv2	2.34	1.014	1.015
MAE	2.60	1.027	1.015

Table 1: Naive vs. controlled far_over_normal ratios averaged across all environments. Naive effects collapse to ≈ 1.0 under latent-distance matching (Controlled); action shuffling has negligible impact (R_{action}).

Figures

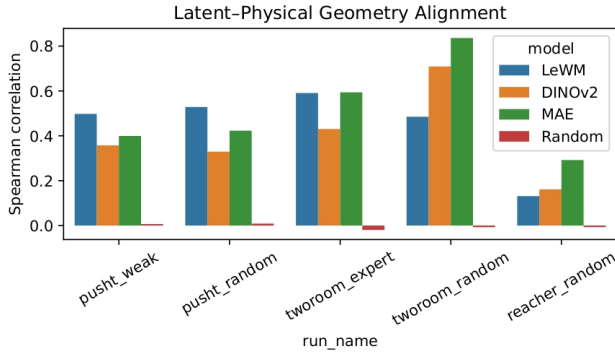


Figure 1: Spearman ρ between latent distance and full physical state distance across models and environments. LeWM shows the strongest alignment in PushT; MAE and DINOv2 dominate in TwoRoom under random policy, where visual position features strongly correlate with ground-truth position. Random baseline is near zero throughout.

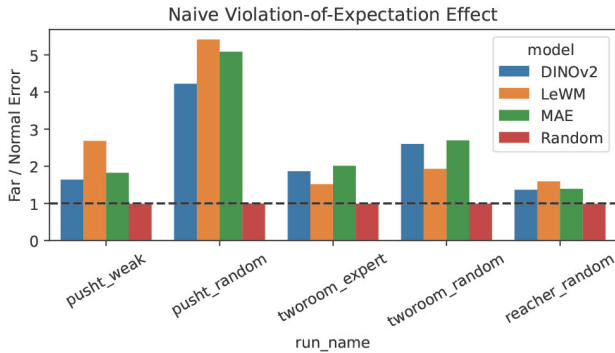


Figure 2: Naive VoE effect (far_over_normal ratio) across models and environments. All learned representations show large apparent surprise for physically distant targets; the random baseline remains at ≈ 1.0 .

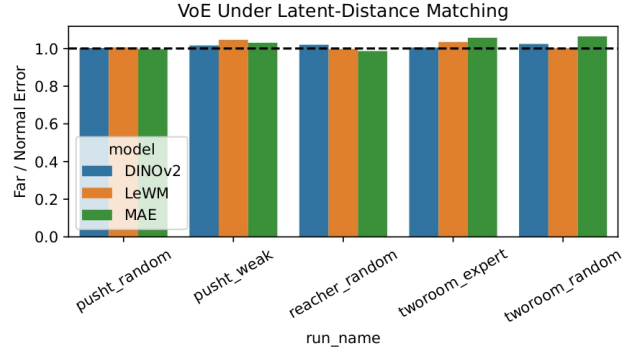


Figure 3: VoE effect under latent-distance matching. The apparent violation signal collapses to between 0.986 and 1.065 across all models and environments compared to naive effects of 1.4x-5.4x.

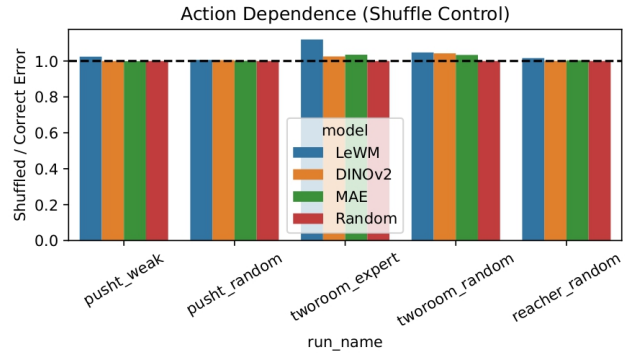


Figure 4: Action sensitivity ratio R_{action} across models and environments. Values near 1 indicate the predictor does not meaningfully use action sequences. All models show $\leq 12\%$ penalty from action shuffling.

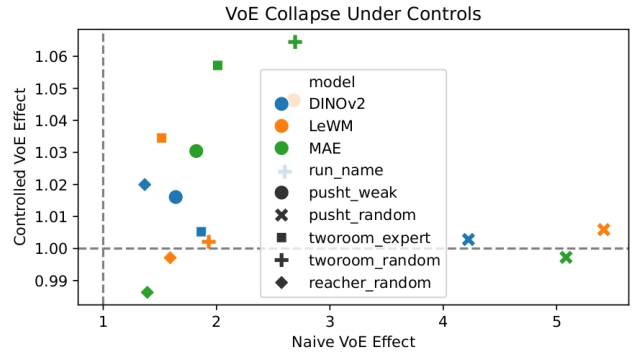


Figure 5: Naive vs. controlled far_over_normal ratios across all models and runs. Points collapse toward 1.0 under the latent-distance control regardless of architecture or environment.

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